

Introduction to Relation

Lecture-6

# Equivalence Relations



Equivalence Relations  
Equivalence Classes  
Equivalence Classes and Partitions

Equivalence Relations

**Definition 1:** A relation on a set  $A$  is called an *equivalence relation* if it is *reflexive*, *symmetric*, and *transitive*.

**Definition 2:** Two elements  $a$ , and  $b$  that are related by an equivalence relation are called *equivalent*. The notation  $a \sim b$  is often used to denote that  $a$  and  $b$  are equivalent elements with respect to a particular equivalence relation.

**Example:** Let  $m$  be an integer with  $m > 1$ . Show that the relation  
$$R = \{(a,b) \mid a \equiv b \pmod{m}\}$$
is an equivalence relation on the set of integers.

**Solution:** Recall that  $a \equiv b \pmod{m}$  if and only if  $m$  divides  $a - b$ .

*Reflexivity:*  $a \equiv a \pmod{m}$  since  $a - a = 0$  is divisible by  $m$  since  $0 = 0 \cdot m$ .

*Symmetry:* Suppose that  $a \equiv b \pmod{m}$ . Then  $a - b$  is divisible by  $m$ , and so  $a - b = km$ , where  $k$  is an integer. It follows that  $b - a = (-k)m$ , so  $b \equiv a \pmod{m}$ .

*Transitivity:* Suppose that  $a \equiv b \pmod{m}$  and  $b \equiv c \pmod{m}$ .

Then  $m$  divides both  $a - b$  and  $b - c$ . Hence, there are integers  $k$  and  $l$  with

$$a - b = km \text{ and } b - c = lm.$$

We obtain by adding the equations:

$$a - c = (a - b) + (b - c) = km + lm = (k + l) m.$$

Therefore,  $a \equiv c \pmod{m}$ .

**Example:** Show that the “divides” relation on the set of positive integers is not an equivalence relation.

**Solution:** The properties of reflexivity, and transitivity do hold, but there relation is not symmetric. Hence, “divides” is not an equivalence relation.

*Reflexivity:*  $a \mid a$  for all  $a$ .

*Not Symmetric:* For example,  $2 \mid 4$ , but  $4 \nmid 2$ . Hence, the relation is not symmetric.

*Transitivity:* Suppose that  $a$  divides  $b$  and  $b$  divides  $c$ . Then there are positive integers  $k$  and  $l$  such that  $b = ak$  and  $c = bl$ . Hence,  $c = a(kl)$ , so  $a$  divides  $c$ . Therefore, the relation is transitive.

Partial Orderings and Partially-ordered Sets  
Lexicographic Orderings  
Hasse Diagrams

Partial Orderings.

**Definition 1:** A relation  $R$  on a set  $S$  is called a *partial ordering*, or *partial order*, if it is **reflexive**, **antisymmetric**, and **transitive**.

A set together with a partial ordering  $R$  is called a *partially ordered set*, or *poset*, and is denoted by  $(S, R)$ . Members of  $S$  are called *elements* of the poset.

Partial Orderings and Partially-ordered Sets  
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Partial Orderings.

**Example 1:** Show that the “greater than or equal” relation ( $\geq$ ) is a partial ordering on the set of integers.

*Reflexivity:*  $a \geq a$  for every integer  $a$ .

*Anti symmetry:* If  $a \geq b$  and  $b \geq a$ , then  $a = b$ .

*Transitivity:* If  $a \geq b$  and  $b \geq c$ , then  $a \geq c$ .

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Partial Orderings.

**Example 2:** Show that the divisibility relation ( $|$ ) is a partial ordering on the set of integers.

*Reflexivity:*  $a \mid a$  for all integers  $a$ .

*Anti symmetry:* If  $a$  and  $b$  are positive integers with  $a \mid b$  and  $b \mid a$ , then  $a = b$ .

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Partial Orderings.

*Transitivity:* Suppose that  $a$  divides  $b$  and  $b$  divides  $c$ . Then there are positive integers  $k$  and  $l$  such that  $b = ak$  and  $c = bl$ . Hence,  $c = a(kl)$ , so  $a$  divides  $c$ . Therefore, the relation is transitive.

$(\mathbb{Z}^+, |)$  is a poset.

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Partial Orderings.

**Example 3:** Show that the inclusion relation ( $\subseteq$ ) is a partial ordering on the power set of a set  $S$ .

*Reflexivity:*  $A \subseteq A$  whenever  $A$  is a subset of  $S$ .

*Anti symmetry:* If  $A$  and  $B$  are positive integers with  $A \subseteq B$  and  $B \subseteq A$ , then  $A = B$ .

*Transitivity:* If  $A \subseteq B$  and  $B \subseteq C$ , then  $A \subseteq C$ .

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Partial Orderings.

**Example 3:** Show that the inclusion relation ( $\subseteq$ ) is a partial ordering on the power set of a set  $S$ .

*Reflexivity:*  $A \subseteq A$  whenever  $A$  is a subset of  $S$ .

*Anti symmetry:* If  $A$  and  $B$  are positive integers with  $A \subseteq B$  and  $B \subseteq A$ , then  $A = B$ .

*Transitivity:* If  $A \subseteq B$  and  $B \subseteq C$ , then  $A \subseteq C$ .

**Definition 2:** The elements  $a$  and  $b$  of a poset  $(S, \preceq)$  are *comparable* if either  $a \preceq b$  or  $b \preceq a$ . When  $a$  and  $b$  are elements of  $S$  so that neither  $a \preceq b$  nor  $b \preceq a$ , then  $a$  and  $b$  are called *incomparable*.

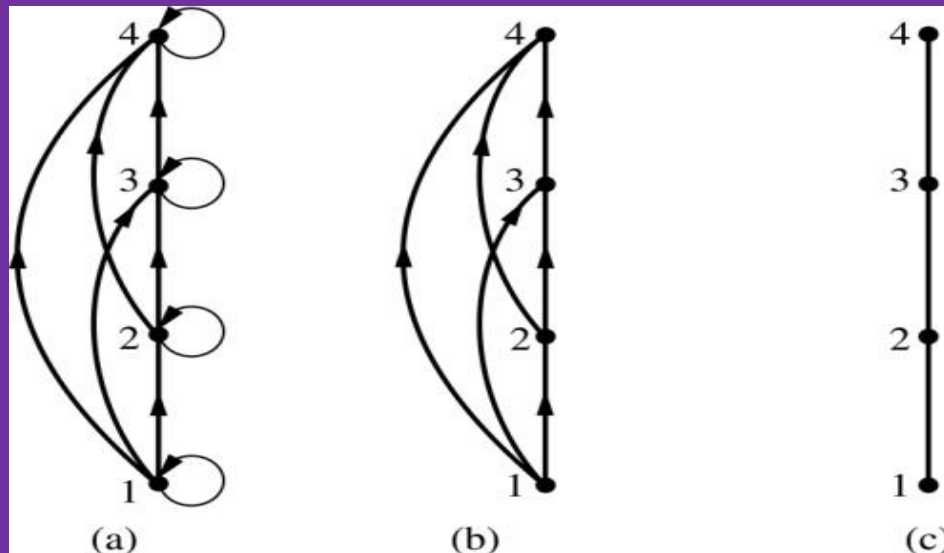
The symbol  $\preceq$  is used to denote the relation in any poset.

**Definition 3:** If  $(S, \preceq)$  is a poset and every two elements of  $S$  are comparable,  $S$  is called a **totally ordered** or *linearly ordered set*, and  $\preceq$  is called a *total order* or a *linear order*. A totally ordered set is also called a *chain*.

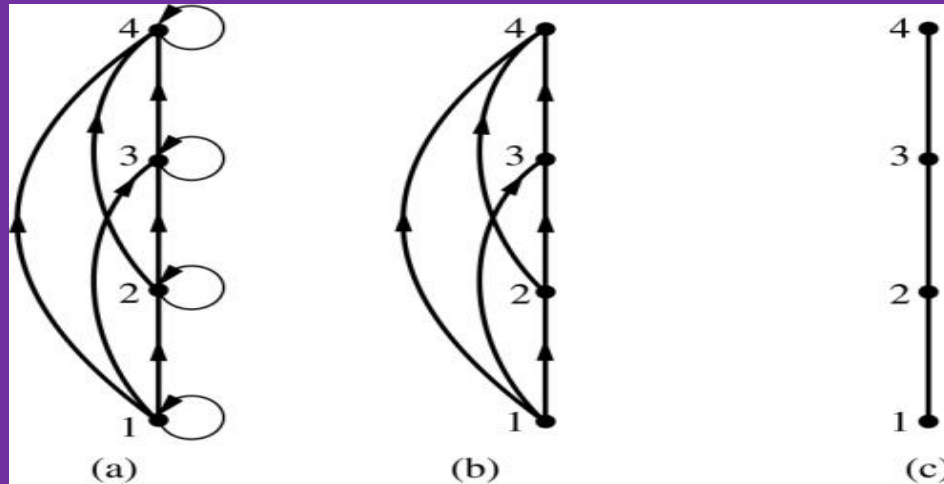
**Definition 4:**  $(S, \preceq)$  is a **well-ordered** set if it is a poset such that  $\preceq$  is a total ordering and every nonempty subset of  $S$  has a least element.

## Hasse Diagrams

**Definition:** A *Hasse diagram* is a visual representation of a partial ordering that leaves out edges that must be present because of the reflexive and transitive properties.



## Hasse Diagrams



A partial ordering is shown in (a) of the figure above. The loops due to the reflexive property are deleted in (b). The edges that must be present due to the transitive property are deleted in (c). The Hasse diagram for the partial ordering (a), is depicted in (c).

## Procedure for Constructing a Hasse Diagram

To represent a finite poset  $(S, \preceq)$  using a Hasse diagram, start with the directed graph of the relation:

- Remove the loops  $(a, a)$  present at every vertex due to the reflexive property.
- Remove all edges  $(x, y)$  for which there is an element  $z \in S$  such that  $x \prec z$  and  $z \prec y$ . These are the edges that must be present due to the transitive property.
- Arrange each edge so that its initial vertex is below the terminal vertex. Remove all the arrows, because all edges point upwards toward their terminal vertex.

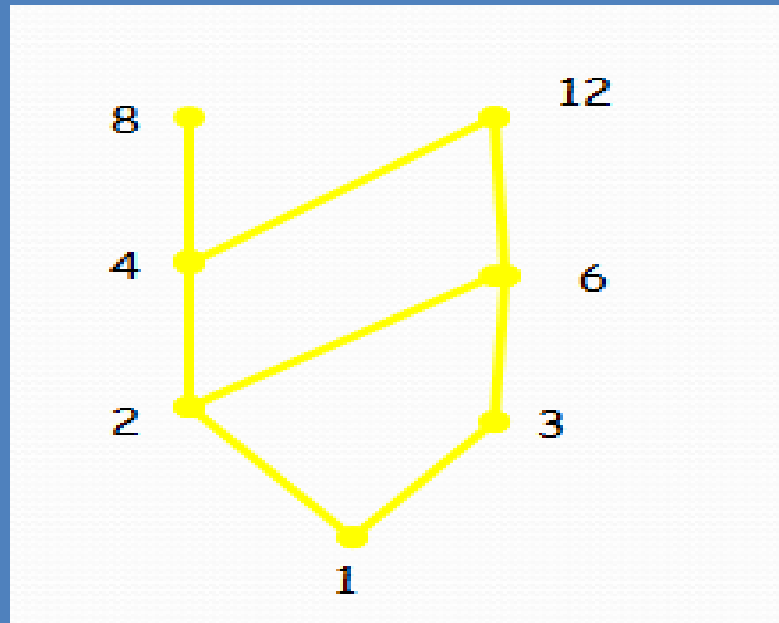
## Procedure for Constructing a Hasse Diagram

Example: Hasse diagram of poset  $(\{1, 2, 3, 4\}, \leq)$



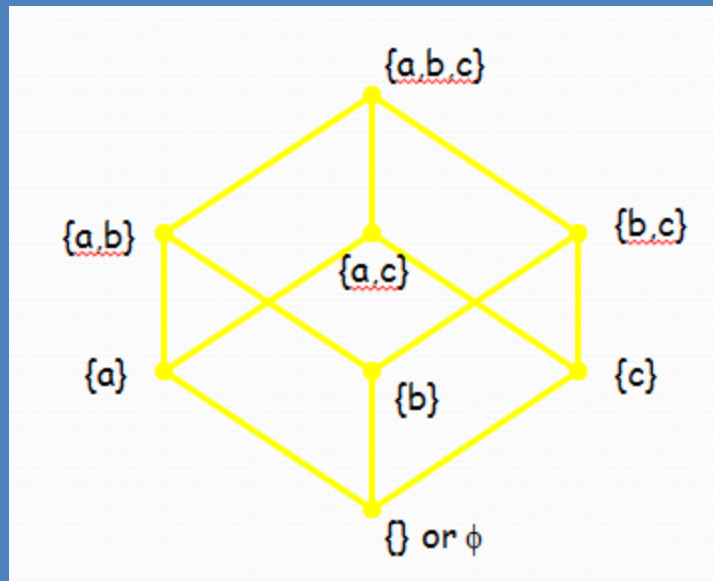
## Procedure for Constructing a Hasse Diagram

Draw the Hasse diagram representing the partial ordering  $\{ (a, b) \mid a \text{ divides } b \}$  on  $\{1, 2, 3, 4, 6, 8, 12\}$



## Procedure for Constructing a Hasse Diagram

Draw the Hasse diagram for the partial ordering  $\{ (A,B) \mid A \subseteq B \}$  on the power set  $P(S)$  where  $S = \{a, b, c\}$



## Minimal and Maximal Elements

Let  $S$  be a partially ordered set.

An element  $a$  in  $S$  is called **minimal** if no other element of  $S$  strictly precedes  $a$ .

An element  $b$  in  $S$  is called **maximal** if no element of  $S$  strictly succeeds  $b$ .

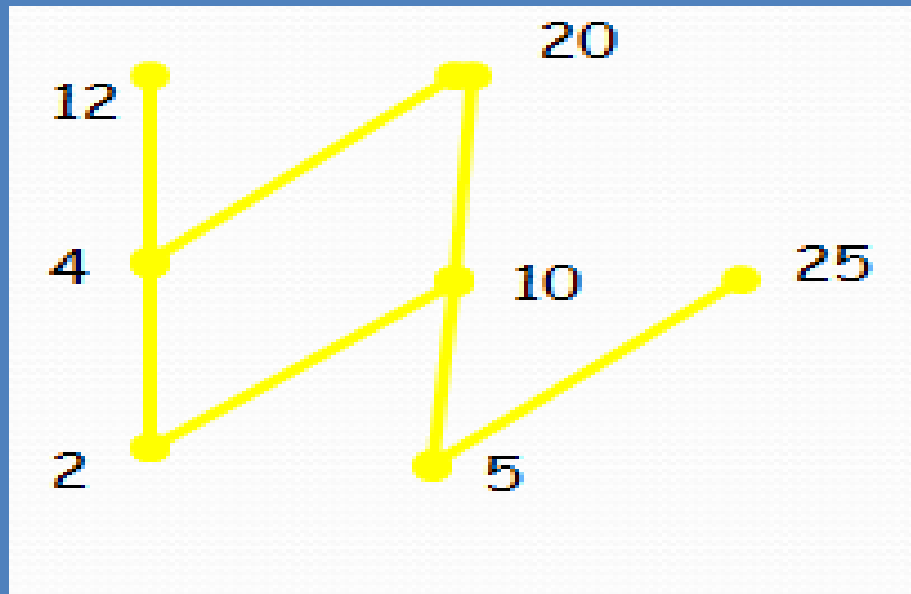
They are respectively *bottom and top* elements in the diagram.

A Poset can have more than one minimal and more than one maximal elements.

## Minimal and Maximal Elements

Which elements of the poset  
 $(\{2, 4, 5, 10, 12, 20, 25\}, |)$  are maximal, and  
which are minimal.

## Minimal and Maximal Elements



The maximal elements are 12, 20 and 25, and the minimal elements are 2 and 5.

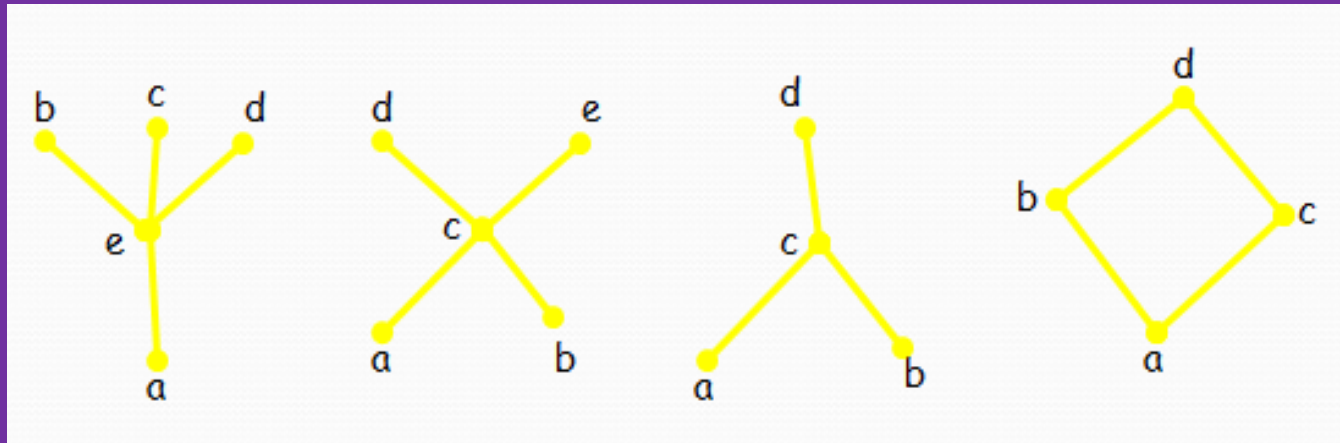
## Minimal and Maximal Elements

An element  $a$  in a poset  $S$  is called least (or first) element if  $a$  precedes every other element of  $S$ .

An element  $b$  in poset  $S$  is called the greatest (or last) element if  $b$  succeeds every other element of  $S$ .

## Minimal and Maximal Elements

Find greatest and least elements in Hasse diagrams



Least element is a no greatest element  
No least element and no greatest element  
No least element and d greatest element  
Least element is a and greatest element d

Note : Greatest and least  
element s are unique  
when it is exist

Let  $A$  be a subset of partially ordered set  $S$ .

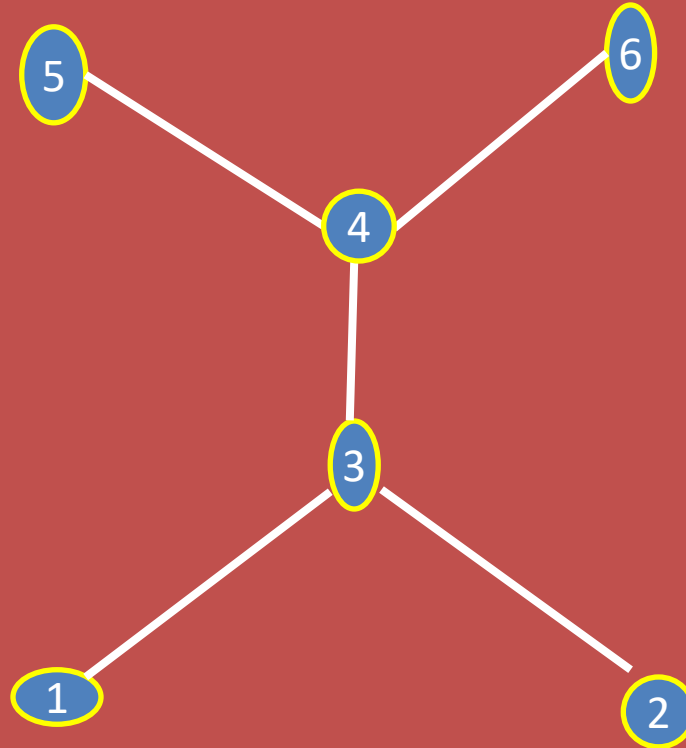
An element  $u$  in  $S$  is called an **upper bound** of  $A$  if  $u$  succeeds every element of  $A$  and  $u$  relate every element of  $A$ .

An element  $l$  in a poset  $S$  is called a **lower bound** of  $A$  if  $l$  precedes every element of  $A$  and  $l$  relate every element of  $A$ .

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## Upper and Lower Bound

What is the upper and lower bound of set  $\{3,4\}$ .  
From following poset



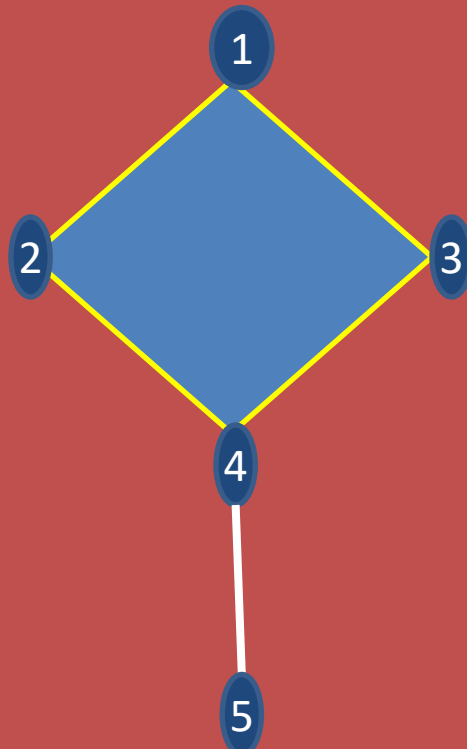
## Upper and Lower Bound

$\{3,4\}$  Is the subset of  $S$ , Now  $3R4, 4R4 \Rightarrow 4$  is upper bound  
 $3R5, 4R5 \Rightarrow 5$  is the upper bound  
 $3R6, 4R6 \Rightarrow 6$  is the upper bound

$\{3,4\}$  Is the subset of  $S$ , Now  $3R3, 4R3 \Rightarrow 3$  is lower bound  
 $3R1, 4R1 \Rightarrow 1$  is the lower bound  
 $3R2, 4R2 \Rightarrow 2$  is the lower bound

## Upper and Lower Bound

What is the upper and lower bound of set  $\{1,3\}$  and  $\{4,5\}$ .  
From following poset of



## Upper and Lower Bound

Lower bound of  $\{1,3\} = \{3,4,5\}$   
Lower bound of  $\{4,5\} = \{5\}$

Upper bound  $\{1,3\}$  is  $\{1\}$   
Upper bound  $\{4,5\}$  is  $\{1,2,3,4\}$

### Supremum

If an upper bound of  $A$  precedes every other upper bound of  $A$ , then it is called the **supremum** or **least upper bound** of  $A$  and is denoted by  **$\sup(A)$**  or  **$\text{lub}(A)$** .

### Infimum

If a lower bound of  $A$  succeeds every other lower bound of  $A$ , then it is called the **infimum** or **greatest lower bound** of  $A$  and is denoted by  **$\inf(A)$**  or  **$\text{glb}(A)$** .

## Minimal and Maximal Elements

Find the lower and upper bounds of the subsets  $\{a, b, c\}$ ,  $\{j, h\}$  and  $\{a, c, d, f\}$  in the poset with the given Hasse diagram.

